

Lab 3

Balancing Act



Physical Models: Torque, Rotational Static Equilibrium, Mass, Weight

Analysis Tools: Propagation of Uncertainty, t-score

Experimental Systems: PASCO Meterstick Balance, Hanging Weights, Mystery Item

Equipment: : Hanging Clips, Bubble Balance, Precision Digital Scale

Safety Concerns: Careful not to allow the meterstick to swing and hit people and, correspondingly, watch out for metersticks while walking around the room!

Introduction

Much of physics is concerned with the scientific investigation of motion. In your prelab you were introduced to the subfield of mechanics called *statics*, the study of conditions under which there is no motion. You were introduced to the concept of *rotational static equilibrium*, and the condition under which it holds for two masses on a pivot ¹ :

$$m_1x_1 = m_2x_2 \quad (3.1)$$

Where m_1 and m_2 denote the masses of the two items and x_1 and x_2 denote their distances from the pivot point, as in the picture below. Our goal in lab this week will be to investigate this model as precisely as possible.

Depending on how you approach the lab, it may be useful for you to instead formulate the relation as:

$$m_1x_1 - m_2x_2 = 0 \quad (3.2)$$

¹As explained in the prelab, this equation comes from $\sum \vec{\tau}_i = 0$, meaning “torques” balance, which is analogous to $\sum \vec{F}_i = 0$ which means all “forces” balance, but they are not the same.

Part 1. A Balancing Act

Your goal in this part of the lab will be to evaluate the validity of this model as precisely as possible. Like last week, you will need to minimize uncertainty as much as possible in order to determine whether the model works. (The model “works” if the mass multiplied by the distance to the mass on one side of the pivot needs to be equal to the mass multiplied by the distance to the mass on the other side of the pivot, in order for the setup to be balanced.)

In previous labs, we learned how to minimize uncertainty by using different equipment and by making multiple measurements. In this lab we’ll encounter a new issue you will need to deal with, which is how uncertainty *propagates* from one variable to another.

Quick Check: Let’s start by configuring the setup. In order to apply the model, the meter stick must be positioned so that the pivot is exactly halfway along it. Move the pivot until the meter stick balances on its own. Is it at the 50 cm mark within uncertainty? Is there any wiggle room in this placement? If there is, is it significant enough to need to be taken into account? Make a decision.

Now add a blue hanger to each end. Note that each of these hangers are 10g! Move them until the meter stick balances exactly. Are they equidistant from the pivot? Based on your findings, does the model seem accurate?

Note: Obtaining a static configuration can be difficult at first, but after practicing it a few times you’ll likely get the hang of it.

Designing Your Experiment: Now, write out a plan for a high-precision test of the model. Some things to consider:

- There are uncertainties in m_1 , x_1 , m_2 , and x_2 . *Hint: You can treat the mass labels as digital measurements.* Depending on how you approach things, you may need to think about how these uncertainties propagate through the model you’re testing. For reference, see the box below.

Extra Guidance: Partial Derivatives and Uncertainty Propagation

Recall that for a single variable $y = f(x)$ uncertainty propagates as:

$$\frac{\delta y}{\delta x} \approx \frac{\partial y}{\partial x} \rightarrow \delta y = f'(x)\delta x \quad (3.3)$$

And that for multiple variables we add individual contributions “in quadrature”:

$$\delta y_{total} = \sqrt{\delta y_1^2 + \delta y_2^2 + \dots} \quad (3.4)$$

Partial derivatives which may be useful for this lab:

$$\frac{\partial}{\partial x} mx = m \quad \frac{\partial}{\partial m} mx = x \quad \frac{\partial m_2}{\partial x_1} = \frac{m_1}{x_2} \quad \frac{\partial m_2}{\partial x_2} = -\frac{m_1 x_1}{x_2^2} \quad \frac{\partial m_2}{\partial m_1} = \frac{x_1}{x_2}$$

- You will need to think about a range of masses and distances to check the model over. As we discovered in the last couple of weeks, just because a model works in a small range, doesn’t mean it works universally! Make sure to try at least 4 configurations of hanging masses. Try at least one configuration in which the masses on either side of the pivot aren’t equal.

- You will need to think about how to minimize uncertainty. given how it propagates (to begin with, you may want to just try a number of points and see how things shake out, but you may want to become more systematic once you get a feel for it. Record choices as you go!)

Check in with Another Group: Briefly discuss your plans with another group. They may provide you with additional ideas and feedback. Note one similarity and one difference between your approaches.

Conducting Your Experiment: Go ahead and give it a shot. You can change your method around while you're at it, you don't need to treat your previous decisions as a contract. Just make sure to record any changes for sake of your organized lab notes. Make sure all of your data is labeled clearly with units, and that you keep track of uncertainty and try to minimize it.

Keep working until you've minimized your uncertainty as much as possible! Treat the denominator of the t-score as the quantity you want to minimize.

Forming a Conclusion: What did you find? What were the limiting factors in reducing the uncertainty? (For example, what were the biggest sources of uncertainty? Is there some measurement you wish you could measure more carefully? Why?) Use this to argue for some possible next steps.

Check in with Another Group: Check back in with the same group you discussed things with before. How do your results compare? Note two similarities and/or differences between what you found, and use this to inform your proposals for future iterations.

Part 2. Beyond the Balancing Act

Based on your results from Part 1, you'll now refine and apply the model to determine the mass of an in-class item. In this part, you'll be able to win a bonus point based on the precision of your results!

Designing Your Experiment: Now that you've assessed the accuracy of the model, we'd like to *apply it*. This is common in science, and represents the transition from hypothesis testing to engineering applications. You may pick any item which you are able to attach to the hangers with the given clip. Your goal will be to predict this object's mass as precisely as possible and then compare it to the object's actual mass. To do this you may want to:

- Decrease the uncertainty in your measurements. You will be allowed to select a *single* mass along with hangers and measure their mass using the precision scale. You can, of course, measure distances using any tool you wish.
- Think carefully about what kinds of configurations minimize the uncertainty that comes from uncertainty propagation.
- Clarify whether there are any regimes where the model fails. You'll want to avoid these or else use them to conclude you cannot determine the mass accurately. (If correct about this you'll still receive the bonus if the model is shown to give the wrong mass when compared with the precision scale)
- Students from previous semesters found it useful to get a rough estimate of the mass of the mystery object in as simple and quick way as possible right away, then working to be more exact and minimize uncertainty after that.

Decide on a plan of action to determine the mass of your item. You'll need to get $t < 1$ when you compare your prediction of the object's mass to its actual mass in order to get full points (or else show that you cannot do this because the model fails in the necessary regime). Whoever gets an accurate result with the *lowest* uncertainty value compared to mass ($\delta m/m$) will get a bonus point!

Conducting Your Experiment: Conduct the experiment as planned. Remember, do everything you can to minimize uncertainty.

When you have your result ready, bring it and your item to your TA for testing. Make sure to do this before you leave lab!

Forming a Conclusion: What did you find? Was your prediction correct? How could you improve your experiment if you were to do it again?

Writing Up Your Lab Notes for Submission

Write up your notes for Part 1 and Part 2 in an organized way according to the rubric provided.